

# ASSESSMENT OF NUTRITIONAL KNOWLEDGE

Nutrition knowledge, broadly defined, refers to knowledge of concepts and processes related to nutrition and health including knowledge of diet and health, diet and disease, foods representing major sources of nutrients, and dietary guidelines and recommendations.

## **Assessment of Nutritional Knowledge**

**Assessment of Nutritional Knowledge** refers to the process of evaluating an individual's understanding of nutrition concepts, healthy eating practices, nutrients, and their role in maintaining health.

### **It helps to determine:**

Awareness about nutrients (carbohydrates, proteins, fats, vitamins, minerals)

Understanding of balanced diet

Knowledge of RDA (Recommended Dietary Allowance)

Awareness of deficiency diseases

Understanding of food sources and healthy food choices

# NATIONAL AND INTERNATIONAL FOOD REGULATORY AGENCIES

A **Food Regulatory Agency** is an official government body responsible for ensuring that food produced, processed, distributed, and sold to the public is **safe, hygienic, and properly labeled**.

## **Need for a Food Regulatory Agency**

A **Food Regulatory Agency** is necessary to ensure that the food consumed by the public is **safe, hygienic, and nutritious**.

1. Protection of Public Health
2. Prevention of Food Adulteration
3. Standardization of Food Quality
4. Regulation of Food Industries
5. Consumer Protection
6. Control of Imported and Exported Foods

# NATIONAL FOOD REGULATORY AGENCY

## FSSAI

Food Safety and Standards Authority of India, established under the Food Safety and Standards Act, 2006, is the regulating body related to food safety and laying down of standards of food in India.

## OBJECTIVES OF FSSAI

The objectives are to protect public health by

- reducing risk of food borne illnesses,
- to protect consumers from unsanitary,
- unwholesome,
- mislabeled or adulterated food.

# INTERNATIONAL FOOD REGULATORY AGENCIES

## **World Health Organization and Food and Agriculture Organization (WHO, FAO)**

The five key principles of food hygiene, according to WHO, are:

- Prevent cross-- contamination of food with pathogens spreading from people, pets, and pests
- Separate raw and cooked foods to prevent contaminating the cooked foods
- Cook foods for the appropriate length of time and at the appropriate temperature to kill pathogens
- Storage of food at the proper temperature
- Use safe water and raw materials

# INTERNATIONAL FOOD REGULATORY AGENCIES

**FAO and WHO** provide expert scientific advice on many aspects of food quality, safety and nutrition relevant to the work of the Codex Alimentarius Commission.

## **Codex Alimentarius**

The Codex Alimentarius is involved in setting up of international food standards, guidelines and codes of practice to contribute to the safety, quality and fairness in international food trade.

Consumers can trust the safety and quality of the food products they buy.

# INTERNATIONAL FOOD REGULATORY AGENCIES

## **International Organization for Standardization (ISO)**

ISO 22000 is a standard developed by the International Organization for Standardization dealing with food safety.

It is a general derivative of ISO 9000.

The ISO 22000 international standard specifies the requirements for a food safety management system that involves the following elements:

- Interactive communication
- System management
- Prerequisite programs

ISO is developing additional standards that are related to ISO 22000. These standards will be known as the ISO 22000 family of standards.

# INTERNATIONAL FOOD REGULATORY AGENCIES

The following standards make up the ISO 22000 family of standards:

- ISO 22000 is about **Food safety management systems** which are requirements for any organization in the food chain.
- ISO 22001 gives the Guidelines on the application of ISO 9001:2000 for the **food and drink industry** (replaces: ISO 15161:2001).
- ISO/TS 22002 deals with the Prerequisite programs on food safety in Food manufacturing
- ISO TS 22003 sets standards for Food safety management systems for bodies providing audit and certification of food safety management systems.
- ISO TS 22004 is for Food safety management systems - Guidance on the application of ISO 22000:2005.
- ISO 22005 gives guidelines to be followed on Traceability in the feed and food chain - General principles and basic requirements for system design and implementation.
- ISO 22006 is regarding Quality management systems - Guidance on the application of ISO 9002:2000 for crop production.
- ISO 22000 is also used in the Food Safety Systems Certification (FSSC) Scheme FS22000. FS22000 is a Global Food Safety Initiative (GFSI) approved scheme.

# INTERNATIONAL FOOD REGULATORY AGENCIES

## **United States-- Food and Drug Administration**

- The FDA Food Safety Modernization Act (FSMA) aims to ensure the U.S. food supply is safe by shifting the focus from responding to contamination to preventing it.
- The USFDA is directly concerned with all aspects covering food safety like recalls, outbreaks & emergencies, safety alerts and advisories, outbreak investigations, and keeping food safe in emergencies.

## **Health Canada**

- Maintaining the safety of Canada's food supply is a shared responsibility among government, industry and consumers.

# NUTRITIONAL ASSESSMENT

Nutritional Assessment is an interpretation of anthropometric, biochemical(laboratory), clinical and dietary survey data to tell whether a person /group of people is well nourished or malnourished(over nourished or under nourished).

There are **direct** and **indirect** methods of assessing Nutritional status.

# DIRECT NUTRITIONAL ASSESSMENT

The direct involve the direct measurement of body dimensions and proportions , determination of tissue or body fluid concentration of nutrients, dietary intake, appearance of the clinical symptoms and signs related to a specific nutrient dependent functional impairment abbreviated as the ABCDs

- A= Anthropometry
- B=Biochemical/Biophysical
- C=Clinical
- D=Dietary

# DIRECT NUTRITIONAL ASSESSMENT

## ANTHROPOMETRIC ASSESSMENTS

Anthropometry comes from two Greek words:

Anthropo = Human, and

Metry/metron = measurement.

Definition: Anthropometry refers to measurement of variations of physical dimension and gross composition of human body at different levels and degrees of nutrition.

Are used to assess the size, shape and composition of the human body

Could be used both in the clinical and field set-ups

# DIRECT NUTRITIONAL ASSESSMENT

## **Purposes of Anthropometric measurements**

Anthropometric measurements are performed with two major purposes:

IN CHILDREN: to assess physical growth.

IN ADULTS: to assess changes in body composition or weight.

# DIRECT NUTRITIONAL ASSESSMENT

## ANTHROPOMETRIC MEASUREMENTS OF GROWTH

- Growth is most rapid in first year of life.
- Children adapt to the chronic nutritional insult by either reducing their rate of growth or by totally failing to grow.
- Therefore Growth performance of children is an excellent reflection of their underlying nutritional status.
- Assessment of growth performance of children is one very important purpose of anthropometric measurements.
- Common anthropometric measurements are weight, height/length, MUAC(Mid-Upper Arm Circumference), head circumference.
- Most accurate and sensitive method of measuring growth is weight gain.

# ANTHROPOMETRIC MEASUREMENTS

## 1. WEIGHT

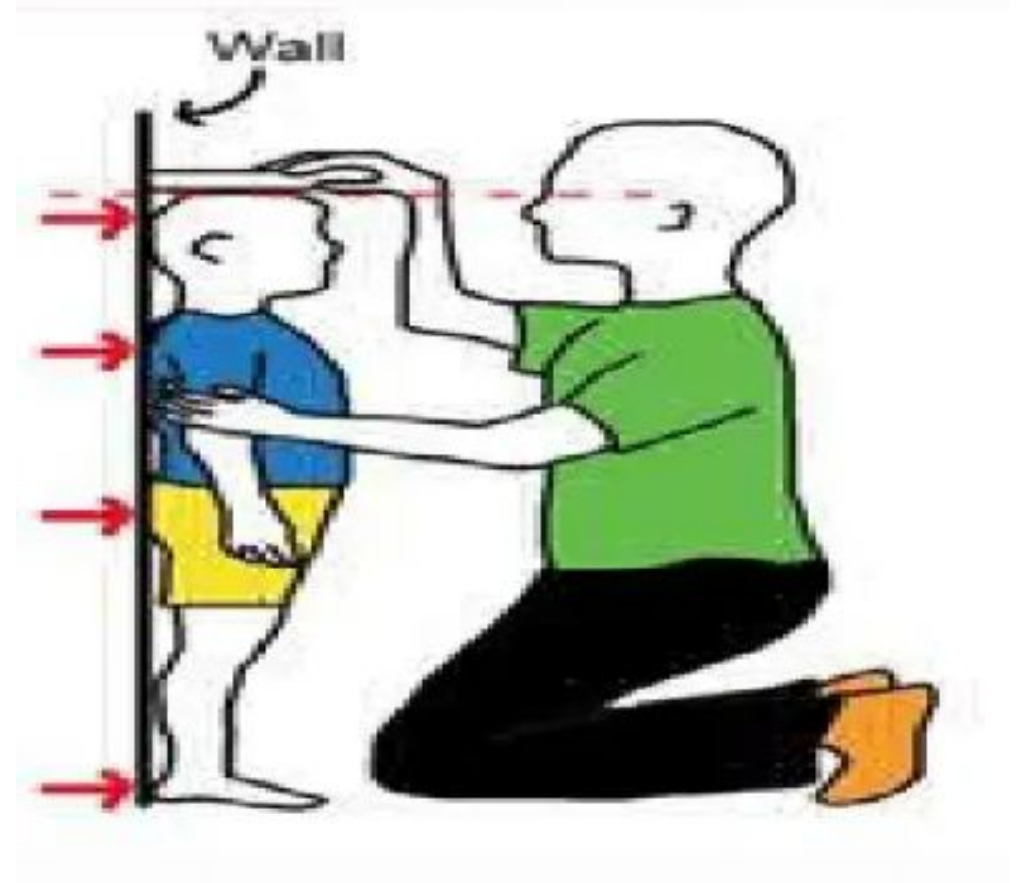
- A weighing sling (spring balance), also called **SALTER SCALE** is used for measuring the weight of the children under 2 years, to the nearest 0.1kg.
- In adults and children over 2 years a beam balance is used and the measurement is also to the nearest 0.1 kg.
- Re-adjust the scale to zero before each weighing.



# ANTHROPOMETRIC MEASUREMENTS

## 2. HEIGHT

- Measured with the adult or child in
- the standing position.
- Either a stadiometer or a portable
- anthropometer can be used for
- measuring.
- Measurements are recorded to the
- nearest millimeter.



# ANTHROPOMETRIC MEASUREMENTS

## HEAD CIRCUMFERENCE (HC)

- Measurement of the head along the supra orbital ridge (forehead) anteriorly and occipital prominence (the prominent area on the back part of the head) posteriorly.
- Measured using flexible , non-stretchable measuring tape around 0.6cm wide.
- HC is useful in assessing chronic nutritional problems in children under 2 years old.
- After 2 years HC is not useful.



# ANTHROPOMETRIC MEASUREMENTS

## MID-UPPER ARM CIRCUMFERENCE

- It is recommended to use a tape to measure the midpoint of the arm between the shoulder and the tip of the elbow.
- MUAC should be measured on only those who are 6 years and below.
- To measure MUAC, a flexible measuring tape is wrapped around the mid-upper arm of the child.
- MUAC should then be measured on the left upper arm while the arm is hanging down the side of the body and relaxed.

MUAC TAPE



# ANTHROPOMETRIC MEASUREMENTS

## SKIN FOLD MEASUREMENT

- A tape is used to measure the midpoint of the arm between the shoulder and tip of the elbow. Fix the caliper at zero before the measurement.
- Skin fold(i.e.,triceps) should be measured at the upper part of the arm.
- Hold the body tissue leaving the muscle. Calliper leaver is released so its spring tension is exerted on the skinfold and maintain pinch while reading the calliper.



# ANTHROPOMETRIC MEASUREMENTS

Measurements that access body composition

- Body Mass Index( $\text{weight} / \text{Square of Height}$ )
- Waist-Hip-Ratio
- Waist-Height-Ratio
- Mid-upper-arm muscle circumference
- Growth indices

# DIET SURVEY

Diet survey is systematic inquiry into the food supplies and food consumption of individual and population groups is made.

## **Needs and Importance of dietary survey**

- Accurate information on dietary patterns of community would help in assessing the nutritional status of the people.
- Helps in determining the relationship between nutritional intake and deficiency diseases.
- Helps in understanding dietary status of the community.

# DIET SURVEY

Commonly used methods are:

1. Family diet survey
  - a. weighment method
  - b. consumption expenditure survey
2. Assessment of dietary intake of individual
  - c. 24 hour recall method
  - d. food record
  - e. Diet history
3. Qualitative Survey
4. Institutional diet survey
5. Food balance sheets

# DIET SURVEY

## 1. Family Diet Survey

Family diet survey collect information on the diet at the household level. The result are expressed as per capita or per consumption unit.

### **Weighment Method**

- This method of diet survey involves actual weighing of raw foods on a given day it is ideal to conduct the survey for 7 days for true picture of the diet.
- In this method every day food is weighted in the morning and evening, before actual cooking. only edible portion of raw food is weighed.
- Information of all the family members who will be consuming the meal on that day is collected according to the age and physiological status.
- Food are converted into nutrients by referring to food consumption tables.
- The nutrient intake expressed as per consumption unit or per capita.
- This method does not give the actual intake of individual in the family.
- This method is accurate as it involves direct weighing of food items.

# DIET SURVEY

## Consumption Expenditure Survey

- In this method ,the money spent on food as well as non food items is assessed by a specially designed questionnaire.
- This method is less cumbersome as it avoids actual weighing of foods.

## 2. Assessment of Dietary Intake Of Individual

### 24 hours recall method

- A 24-hour dietary recall (24HR) is a structured interview intended to capture detailed information about all foods and beverages eaten by a respondent in the past 24 hours.
- This can be done via a face-to-face interview or telephone.
- The interview is often structured, with a standardized and pretested protocol involving probing questions that help the respondent remember all the items consumed and relevant food and beverage descriptions.
- Information on food type, preparation methods, recipes, brand names of commercial products, and estimated portion size are usually required.

# DIET SURVEY

## **Food Record**

Food record or dairy method provides information about individuals. In this method, the subject records , at the time of consumption, the type and amount of all food and drinks consumed for a particular period of time , usually from 1 to 7 days.

The strength of this method is that doesn't depend upon memory and provide detailed food intake data and also about eating habits.

## **Diet History**

- A method used to assess a person's usual long-term dietary intake over weeks or months.
- Information is collected through a detailed personal interview about daily eating habits.
- Records meal pattern, types of foods consumed, and frequency of intake.
- Includes details about portion sizes and cooking methods.
- Helps to identify nutritional deficiencies or excess intake.
- Useful for diet planning, counseling, and research purposes.

# DIET SURVEY

## 3. Qualitative Diet Survey

### **Food Frequency Method**

- This method is consists of asking individuals how often daily, monthly, weekly specific foods are eaten.
- Food Frequency questionnaire are relatively inexpensive and quick to administer in large scale surveys.
- They are also considered one of the methods of choice for research on diet - diseases relationship on both macro and micro nutrient.
- Limitation is that food list is limited to 100 or fewer food or food groups.

# DIET SURVEY

## 4. Institutional Diet Survey

- This method is often employed in institutions like hostels, army barracks, orphanages and homes for the aged, where homogeneous groups of people take their meal from common kitchen
- In this method , the amount of food stuff issued to kitchen as per the records are taken into the consideration.
- No direct measurement is done .
- Average intake of per person per day=
$$\frac{\text{stock at the beginning of the week} - \text{stock at the end of the week}}{\text{total no. of inmates} \times 7}$$

# DIET SURVEY

## 5. Food Balance Sheet

- Food balance sheets (FBS) from the Food and Agriculture Organization of the United Nations (FAO) are used to assess the national availability of food for consumption.
- They are widely used in the food and agriculture sectors to monitor global food and dietary patterns.
- FAO FBS only provide data on annual per capita consumption of energy, protein, and fat (not micronutrients).
- FBS have also been used to set public health priorities, formulate policies, undertake inter-country comparisons, monitor trends over time, and estimate the likelihood of micronutrient deficiencies.

# INDIRECT NUTRITIONAL ASSESSMENT

Indirect method includes assessment of indicators of the food and nutrition situations in the area/region of interest by looking at certain data that are closely related to malnutrition or which are aggravated by malnutrition.

These include

- Cause specific mortality rates
- Age specific mortality rates
- Health service statistics
- Rate of nutritionally relevant infections
- Meteorological data
- Income level
- Market price of foods
- Predominance of cash crops